

Md Minhazul Islam

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About Me

I have recently completed my BSc in Computer Science and Engineering from Rajshahi University of Engineering and Technology. My research interests center on large language models (LLMs), multimodal analysis, agentic AI, retrieval-augmented generation (RAG), natural language processing, and medical text analysis. I am passionate about advancing AI-driven systems and have contributed to several research projects and publications in these areas.

Research Interests

- Natural Language Processing (NLP), Computational Linguistics, and Low-Resource Language Processing
- Agentic AI-based system implementation with RAG
- Medical Text Analysis, Healthcare Informatics, and Semantic Search
- Large Language Models (LLMs), Multimodal Learning, and Contextual Embeddings
- Explainable AI, Model Interpretability, and Topic Modeling

Publications

Journal Articles

- [J1] Rashid, M. H. O., Jubaer, M. T., **Islam, M. M.**, et al. (2024). *Live and Mediated User Engagements: A Comparative Dataset from Two Bengali Audio-Story-Based YouTube Channels*. *Data in Brief, Elsevier*. [Data Translation, Revision]

Conference Papers

- [C1] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y. (2023). *Navigating Bengali Linguistics: Insights from Machine and Deep Learning Perspectives for Categorization of Sentences*. *ICCIT 2023*. [Code, Methodology]
- [C2] **Islam, M. M.**, Jubaer, M. T., Rashid, M. H. O. (2024). *Consistency of Contextual Embedding in Literary Texts*. *ICISSET 2024*. [Main Idea, Code, Methodology]
- [C3] Jubaer, M. T., Chowdhury, B., **Islam, M. M.**, Rashid, M. H. O. (2024). *Detecting Discrepancy to Enhance the Accuracy of Sentiment and Emotion Analysis*. *STI 2024*. [Coding, Abstract]
- [C4] Jubaer, M. T., Farzana, M., Chowdhury, B., Zaman, M. S. U., Srizon, A. Y., **Islam, M. M.** (2024). *Analyzing Audience Engagement in Esports: Sentiment and LLM-Based Topic Insights*. *ICCIT 2024*. [Literature, Revision]
- [C5] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y., Hossain, M. A. (2024). *Explainable AI-Driven Improved Disease Prediction through Symptom Analysis*. *ICCIT 2024*. [Idea, Code, Methodology, Conclusion]

Experience

- **AI Engineer** June 2025 – September 2025
makebell Limited, Hong Kong (makebell.com) Remote
 - Developing and deploying advanced AI-powered applications tailored for the legal industry, applying RAG with Agentic AI to build a domain-specific translation system.
 - Specializing in prompt engineering, large language model (LLM) integration, and optimizing backend architectures for scalable AI services.
 - Contributing to real-time, high-impact product development in a fast-paced and competitive environment.
- **Young Learners Research Lab (Research Assistant)** Jan 2023 – Present
On-site
 - Conducting research on NLP and deep learning, focusing on enhancing model efficiency and explainability.
 - Working with Bengali and English datasets, we address underexplored linguistic domains.

- Coauthor of multiple conference and journal publications, with ongoing research projects.
- **Computational Linguistics Research Lab (Research Assistant)** Jun 2023 – Present
Remote
 - Investigating word embeddings, contextual embeddings, and topic modeling in literary texts.
 - Analyze linguistic patterns in American slave narratives and perform stylometric analysis.
 - Developing novel NLP techniques for literary research and computational linguistics applications.
- **Outlier (AI Trainer — Mathematics and Coding, Contributor & Quality Control)** June 2024 – Present
Remote
 - Contributing to dataset creation and refining AI model outputs in coding and mathematics.
 - Performing rigorous quality control on AI-generated content to ensure accuracy and reliability.
- **Talent IT and ICT (Teacher — ICT Instructor)** 2020 – 2022
Rajshahi, Bangladesh
 - Instructing students in Information and Communication Technology (ICT).
 - Designing and delivering engaging lessons for secondary and higher secondary levels.

Education

- **Rajshahi University of Engineering and Technology** Jan 2020 – August 2025
Rajshahi, Bangladesh
 - BSc in Computer Science and Engineering
 - CGPA: 3.42/4.00
- **Rajshahi Govt. City College** May 2017 – Apr 2019
Rajshahi, Bangladesh
 - Higher Secondary Certificate
 - GPA: 5.00/5.00

Projects

- **Contextual Sarcasm Detection (Undergraduate Thesis)** 2025

 - Python, PyTorch, Transformers, Pandas, NumPy
 - * Developed an encoder-encoder cross-attention model for contextual sarcasm detection.
 - * Combined CLIP for image encoding and mBERT for text processing, achieving 95% accuracy.
 - * Currently refining a custom multimodal architecture for enhanced performance.
- **Clickbait Content Detection** 2023–2024

 - Python, Scikit-learn, PyTorch, Transformers, CLIP, Pandas, NumPy
 - * Built a tool to detect clickbait on social media platforms using multimodal analysis.
 - * Integrated CLIP and mBERT, achieving 95% accuracy in initial evaluations.
 - * Ongoing data collection and model refinement for improved reliability.
- **Medical Triage Classification** 2024–2025

 - Python, Transformers, Scikit-learn, Pandas, NumPy, Matplotlib
 - * Developed a custom BERT-based model for classifying medical triage notes.
 - * Designed regex and medical dictionary for text preprocessing, achieving 83% accuracy.
 - * Explored multimodal and EDBERT models for enhanced classification performance.
- **US Election Data Analysis** 2025

 - Python, PyTorch, LLM, PCA, Topic Modeling, Transfer Learning
 - * Analyzed Twitter posts using topic modeling, BERT-based semantic search, and GraphRAG.
 - * Clustered embeddings with t-SNE and HDBSCAN to extract election-related insights.
 - * Developed an alternative to traditional exit polls, with ongoing evaluation.

Technical Skills

Languages: Python, C, C++, PHP

ML/DL Frameworks: PyTorch, Scikit-learn, Transformers, BERT, RAG, Agno, LoRA, Quantized LLM Finetuning

Tools & Libraries: Matplotlib, AWS, Prompt Engineering

Honors and Awards

- **Dataverse Challenge (Top 15 Finalist)** 2023
IT Verse 
 - * Ranked in the Top 15, demonstrating strong data analysis and machine learning skills.
- **CUET ETE DAY 2023 ML Competition (Top 10 Finalist)** 2023
CUET 
 - * Secured a Top 10 position, showcasing expertise in model building, evaluation, and optimization.
- **Robi Datathon 3.0 (Finalist)** May 2024
Robi Headquarters, Dhaka 
 - * Reached the final round, exhibiting strong data science, analytics, and problem-solving skills.

Leadership Experience

- **Officer Cadet** Jan 2022 – Apr 2022
Bangladesh Military Academy 
 - * Completed a 3-month training program in military discipline, tactics, and leadership.
 - * Developed skills in teamwork, resilience, and time management under pressure.
- **Team Leader - Robi Datathon 3.0 Finalist** May 2024
Robi Headquarters, Dhaka 
 - * Led a team to the final round, developing a data-driven business solution.
 - * Coordinated efforts to ensure efficient problem-solving and innovation.

Volunteer Experience

- **Social Worker** 2024 – Present
BD Clean Bagmara, Rajshahi 
 - * Working as an IT advisor and active member.
 - * Raised awareness on environmental conservation, cleanliness, and waste management.
 - * Participated in community events to promote health and hygiene practices.
- **AI Research Mentor** 2023 – Present
University Research Community 
 - * Mentored students in AI, ML, and data analysis, organizing knowledge-sharing sessions.

Certifications

- **Green Card – Eligible for Military Officer Course** Dec 2021
Inter Services Selection Board, Bangladesh
- **Certificate of Appreciation - 26th Int. Conf. On Computer and Information Technology** Organizer: IEEE Bangladesh Section
2023
- **Conference Paper Presentation (ICISSET)** 2024
International Islamic University Chittagong
- **Conference Paper Presentation (ICCIT-2024)** Dec 2024
IEEE Bangladesh Section

Language Proficiency

Bengali: Native (Speaking, Listening, Reading, Writing)
English: Intermediate (Speaking: Intermediate, Listening: Intermediate, Reading: Upper-Intermediate, Writing: Intermediate)

Hobbies and Interests

Interests: Football, Travel, Fitness, Watching adventure and war movies

References

1. **Dr. Md. Ali Hossain**
Professor and Head, Department of Computer Science and Engineering, RUET
Email: ali.ruet@gmail.com
Relationship: Research Supervisor – Medical Text Analysis
2. **Mohammad Harun Or Rashid (PhD Student)**
Graduate Research Assistant, Department of English, University of Louisiana at Lafayette
Assistant Professor (On Study Leave), Department of English, RUET
Email: harunmr9@gmail.com
Relationship: Research Supervisor (Text Analytics, Digital Humanities)
3. **Md. Azmain Yakin Srizon**
Assistant Professor, Department of Computer Science and Engineering, RUET
Email: azmainsrizon@cse.ruet.ac.bd
Relationship: Undergraduate Thesis Supervisor, Research Supervisor (NLP and LLM)